Introduction to Mathematics and Optimization

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The references in the suggested solutions to this examination paper are with respect to the PDF edition *Introduction to Mathematics and Optimization* (Compressed static version (26.11.2020) of interactive book).

PROBLEM 1

Let $f : U \to \mathbb{R}$ be the function given by

$$f(x, y) = x^3 + 2y^3,$$

where

$$U = \{(x, y) \in \mathbb{R}^2 \mid x < 1,\ y < 1,\ x + y > 1\}$$

(a) Let $S = \mathbb{R}^2 \setminus U$. Write S using weak inequalities (i.e., inequalities with $\leq$ and $\geq$). Show that S is a union of closed sets.

ANSWER:

The complement $\mathbb{R}^2 \setminus U$ of U is given by

$$S = \{(x, y) \in \mathbb{R}^2 \mid x \geq 1 \ \vee\ y \geq 1 \ \vee\ x + y \leq 1\}.$$

Therefore S can be written as the union

$$\{(x, y) \mid x \geq 1\} \cup \{(x, y) \mid y \geq 1\} \cup \{(x, y) \mid x + y \leq 1\}.$$

Each of these three sets is closed because they are preimages of closed sets under continuous functions. Here we use Proposition 5.42 and Proposition 5.44. For example,

$$\{(x, y) \mid x + y \leq 1\} = \{(x, y) \mid g(x, y) \in (-\infty, 1]\} = g^{-1}((-\infty, 1]),$$

where $g(x, y) = x + y$.

(b) Show that f is a strictly convex function. It may be used without proof that U is an open convex set.

ANSWER:

The Hessian matrix of f is

$$\nabla^2(f) = \begin{pmatrix} 6x & 0 \\ 0 & 12y \end{pmatrix}.$$

When $(x, y) \in U$, we have $x > 0$ and $y > 0$. For example, $x < 1$ and $x + y > 1$ imply that $x < x + y$ and hence $y > 0$.

Therefore the Hessian matrix is positive definite according to Exercise 8.22 for $(x, y) \in U$, and the function is strictly convex by Theorem 8.17.

Consider the optimization problem

$$\begin{aligned} &\min f(x, y) \\ &(x, y) \in C, \end{aligned} \tag{1}$$

where

$$C = \{(x, y) \in \mathbb{R}^2 \mid x \le 1,\, y \le 1,\, x + y \ge 1\}.$$

(c) Explain, by a simple reference to the curriculum without calculations, why there must exist an optimal solution to (1).

ANSWER:

The function f is continuous and C is closed and bounded for the same reasons as in the solutions to previous examination papers. Therefore an optimal solution exists by Theorem 5.51.

(d) Solve the optimization problem (1) using a method from the curriculum. Is the solution unique?

ANSWER:

The solution must lie on the boundary by Proposition 7.38, since f has no critical points in C. In fact, the only global critical point is $x = 0$ and $y = 0$. We first check $x = 1$ and must then minimize $1 + 2y^3$ for $0 \le y \le 1$. This gives a minimum of 1 for $y = 0$ (and $x = 1$). For $y = 1$, we must minimize $x^3 + 2$ for $0 \le x \le 1$. This gives a minimum of 2 for $x = 0$ (and $y = 1$). For $x + y = 1$, we must minimize $g(x) = x^3 + 2(1 - x)^3$ for $0 \le x \le 1$. We differentiate and set equal to 0: $3x^2 - 6(1 - x)^2 = -3x^2 + 12x - 6 = 0$, which gives $x = 2 - \sqrt{2}$ (the other root is $2 + \sqrt{2} > 1$) and hence $y = \sqrt{2} - 1$. We obtain numerically that $g(2 - \sqrt{2}) = 2(3 - 2\sqrt{2}) = 0.34\ldots$. This is our minimal value, because $g(0) = 2$ and $g(1) = 1$. That is, the optimal solution is $x = 2 - \sqrt{2}$ and $y = \sqrt{2} - 1$. The solution is unique, since our method above shows that a solution (x, y) must satisfy $-3x^2 + 12x - 6 = 0$. This quadratic equation has the two solutions $2 - \sqrt{2}$ and $2 + \sqrt{2}$. Since $2 + \sqrt{2} > 1$, only the first solution can be used, given that $x \le 1$ and $1 = x + y$. This uniquely determines both x and y.

PROBLEM 2

Let

$$f(x,y,z) = x^2 + 2y^2 + z^2 - 2xy + 2yz$$

and

$$v = \begin{pmatrix} x \\ y \\ z \end{pmatrix}.$$

(a) Let

$$B = \begin{pmatrix} a & d & 0 \\ d & b & e \\ 0 & e & c \end{pmatrix},$$

where $a, b, c, d, e \in \mathbb{R}$. Show that

$$v^T B v = ax^2 + by^2 + cz^2 + 2dxy + 2eyz.$$

ANSWER:

Via matrix multiplication we obtain

$$\begin{aligned} v^T B v &= \begin{pmatrix} x & y & z \end{pmatrix} \begin{pmatrix} a & d & 0 \\ d & b & e \\ 0 & e & c \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x & y & z \end{pmatrix} \begin{pmatrix} ax + dy \\ dx + by + ey \\ ey + cz \end{pmatrix} \\ &= x(ax + dy) + y(dx + by + ey) + z(ey + cz) = ax^2 + by^2 + cz^2 + 2dxy + 2eyz. \end{aligned}$$

(b) Use (a) to find a symmetric 3×3 matrix A such that

$$\begin{aligned} f(x,y,z) &= x^2 + 2y^2 + z^2 - 2xy + 2yz \\ &= v^T A v. \end{aligned}$$

ANSWER:

From (a) we see that

$$A = \begin{pmatrix} 1 & -1 & 0 \\ -1 & 2 & 1 \\ 0 & 1 & 1 \end{pmatrix},$$

since $a = 1, b = 2, c = 1, 2d = -2$ and $2e = 2$.

(c) Show that $f(x,y,z) \geq 0$ for all $x, y, z \in \mathbb{R}$.

ANSWER:

This is equivalent to showing that the matrix A in the previous part is positive semidefinite. We use the procedure in Example 8.24 (based on Theorem 8.23). Below only the results of the operations on rows and columns are indicated.

$$\begin{pmatrix} 1 & -1 & 0 \\ -1 & 2 & 1 \\ 0 & 1 & 1 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

The last matrix is a diagonal matrix with non-negative entries on the diagonal. It is therefore positive semidefinite according to Exercise 8.22. We have thus shown that A is positive semidefinite.

(d) Find $x, y, z \in \mathbb{R}$ not all zero, such that $f(x, y, z) = 0$. Explain your method.

ANSWER:

Starting from the matrix A, a $v \neq 0$ with $Av = 0$ will give the answer. This is equivalent to solving the equations

$$\begin{aligned} x - y &= 0 \\ -x + 2y + z &= 0 \\ y + z &= 0. \end{aligned}$$

A solution could be (verified by CAS): $x = y = -1$ and $z = 1$.

(e) Is f a strictly convex function?

(**hint**: Consider $u = 0 \in \mathbb{R}^3$ and $v \in \mathbb{R}^3$ with $v \neq 0$ and $f(v) = 0$. It may be used without proof that $f(tv) = t^2 f(v)$ for $t \in \mathbb{R}$)

ANSWER:

Set $u = 0$ and let $v \neq 0$ satisfy $f(v) = 0$. Then

$$f((1-t)u + tv) = f(tv) = t^2 f(v) = 0 = (1-t)f(u) + tf(v).$$

Therefore the function cannot be strictly convex according to Definition 6.1.

PROBLEM 3

Let $f : \mathbb{R}^3 \to \mathbb{R}$ be the function given by

$$f(x, y, z) = x^2 + y^2 + z^3 + xy + x + y - z.$$

(a) Find the critical points of f.

ANSWER:

The critical points are given as solutions to $\nabla f(v) = 0$, i.e., as solutions to the equations

$$\begin{aligned} 2x + y + 1 &= 0 \\ x + 2y + 1 &= 0 \\ 3z^2 - 1 &= 0. \end{aligned}$$

Here there are two solutions (found via CAS):

$$x = y = -\frac{1}{3}, \qquad z = \pm\frac{1}{\sqrt{3}}.$$

(b) Determine for each of the critical points whether they are local maxima, minima, or saddle points.

ANSWER:

The Hessian matrix of f at the point (x, y, z) is

$$\nabla^2(f) = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 2 & 0 \\ 0 & 0 & 6z \end{pmatrix}.$$

By operations as indicated in Example 8.24 we obtain

$$\begin{pmatrix} 2 & 1 & 0 \\ 1 & 2 & 0 \\ 0 & 0 & 6z \end{pmatrix} \sim \begin{pmatrix} 2 & 0 & 0 \\ 0 & \frac{3}{2} & 0 \\ 0 & 0 & 6z \end{pmatrix}.$$

Therefore the Hessian matrix at a point (x, y, z) is positive definite for $z > 0$ and indefinite for $z < 0$ (in particular not positive semidefinite). This follows from Exercise 8.22. From Theorem 8.9 one can therefore conclude that

$$\left(-\frac{1}{3}, -\frac{1}{3}, \frac{1}{\sqrt{3}}\right)$$

is a local minimum, while

$$\left(-\frac{1}{3}, -\frac{1}{3}, -\frac{1}{\sqrt{3}}\right)$$

is a saddle point.

(c) Is f a convex function?

ANSWER:

Since $\nabla^2(f)$ is not positive semidefinite for $z < 0$, f cannot be convex. This follows from Theorem 8.17.

(d) Give an open convex set $U \subseteq \mathbb{R}^3$ such that f considered as a function on U is strictly convex.

ANSWER:

On $U = \{(x, y, z) \mid z > 0\}$, f is a convex function by Theorem 8.17, because $\nabla^2(f)$ is positive definite there. Here U is an open set because the complement $\mathbb{R}^3 \setminus U = \{(x, y, z) \mid z \leq 0\}$ is closed. This is seen from the fact that $g(x, y, z) = z$ is continuous, and $\mathbb{R}^3 \setminus U = g^{-1}((-\infty, 0])$ via Proposition 5.44 and Proposition 5.42.